Week 12

Summative Assignment

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DS-7003 Advanced Decision Making: Predictive Analytics & Machine Learning

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Abstract

Online retailers generate continuous streams of clickstream data, yet many organizations struggle to turn this information into timely, actionable decisions. This report analyzes 12,330 e-commerce browsing sessions to identify the behavioral patterns that precede a purchase and to build a model that predicts purchasing intent. Following the CRISP-DM methodology, exploratory analysis was used to examine the data and seven classification algorithms (Logistic Regression, Decision Tree, Random Forest, AdaBoost, Gradient Boosting, Naive Bayes and a Multilayer Perceptron) were compared. Gradient Boosting performed best overall (AUC 0.935, recall 0.855 on the test set), although cross-validation showed that it ranks sessions no better than the Random Forest, the two differing mainly in their balance of recall and precision. Page Value was by far the strongest predictor, and conversion also varied by month and visitor type. The report further shows that the classification threshold should be set according to the cost and benefit of the intervention rather than left at the default 0.5. The findings are used to derive operational and marketing recommendations for reducing abandonment and improving conversion.

All analysis code, including data preparation, model training, cross-validation and threshold analysis, is available at https://github.com/AtomHarris/E-Commerce-Purchase-Intention.

1. Introduction

E-commerce organizations accumulate vast quantities of behavioral data through website visits, yet its value depends on an organization's ability to translate it into decisions. However, translating raw data into decisions is not automatic and requires a structured process such as the widely adopted CRISP-DM framework, which sequences data mining projects through business understanding, data preparation, modelling and deployment phases (Chapman et al., 2000).

The organization examined in this report is an e-commerce company that has experienced frequent operational breakdowns, unplanned interruptions and unexpected costs, resulting in lost revenue and reduced customer satisfaction. The company has therefore adopted a connected analytics platform to better understand its operational performance, but currently lacks a systematic method for turning session-level browsing data into forward-looking decisions. This gap gives rise to two research questions:

I. What patterns in customer browsing behavior distinguish sessions that end in a purchase from those that do not?

II. Can a predictive model be built that reliably classifies a session as likely or unlikely to convert, and can its output support timely decisions such as targeted offers, live-chat intervention, or dynamic content?

These questions matter because even a marginal improvement in early purchase-intent prediction allows interventions to be targeted at genuinely high-intent sessions rather than applied indiscriminately. A related question concerns how probabilistic predictions should be converted into operational decisions. Sections 2.4 and 3.5 address this by considering the costs and benefits of acting on a prediction rather than treating it as certainty.

1.1 Objectives

The main objectives of this report are:

i. To review the relevant literature on predictive analytics, decision-making theory and machine learning in e-commerce.

ii. To explore and describe the structure of the dataset, including its architecture and attributes.

iii. To build, evaluate and compare multiple predictive classification models using Python.

iv. To assess the accuracy, uncertainty and business utility of the resulting predictions

v. To translate the statistical findings into practical recommendations for the organization's decision-making process.

The remainder of this report reviews the literature, dataset architecture, modelling techniques and mathematical foundations, before presenting the analysis, model comparison, error analysis, uncertainty assessment and recommendations.

2. Literature Review

2.1 Predictive Analytics and Decision-Making in E-Commerce

According to Shmueli and Koppius (2011), predictive analytics refers to statistical and machine learning techniques used to estimate future outcomes from historical data. In an e-commerce context, predictive analytics enables an organization to move from reactive to proactive decision-making. Rather than reviewing sales figures after the fact, an organization can anticipate which visitors are likely to convert, abandon their sessions, or interact with commercially important pages. The same approach can support operational-failure prediction, customer lifetime-value estimation and churn-risk identification.

This shift supports three broad decision-making functions identified in the assessment brief by filtering noise from large volumes of session data, uncovering relationships between behavioral variables, and directing marketing investment towards channels and periods most likely to generate returns.

A structured process is essential to ensure that predictive analytics projects deliver decision-relevant outcomes rather than purely academic exercises. The Cross-Industry Standard Process for Data Mining (CRISP-DM) provides a widely adopted six-phase framework comprising business understanding, data understanding, data preparation, modelling, evaluation and deployment (Chapman et al., 2000). This report broadly follows this structure where Sections 2 and 3 cover business and data understanding, preparation, modelling and evaluation, while Section 4 addresses deployment oriented recommendations.

In analytics, predictive and descriptive analytics are both important, but predictive analytics is often confused with descriptive reporting. A dashboard showing last month's bounce rate or total sessions is descriptive because it reports what has already happened. Predictive analytics goes further by estimating what a customer is likely to do while a session is still in progress (Provost and Fawcett, 2013). Descriptive metrics can therefore support decisions after an event, whereas a predictive score can support intervention before the outcome is known. This distinction directly addresses the case study organization's need to “recognise and identify performance patterns” and “forecast and predict unsolicited events”.

2.2 System Architecture and the Dataset

The case-study organization's architecture is typical of a modern e-commerce platform, consisting of a front-end website that captures page-level interactions, a backend that logs session activity, and a reporting layer that presents descriptive summaries such as total visits or exit rates. The gap addressed by this report is the absence of a predictive layer that scores each session according to its likelihood of resulting in a purchase. This architecture is consistent with the real-time behavior analysis system proposed by Sakar et al. (2019), in which page-view and session data are supplied to classifiers that predict purchasing intention and site abandonment.

The dataset used in this study is the Online Shoppers Purchasing Intention Dataset, comprising 12,330 website sessions collected over roughly a one-year period (Sakar and Kastro, 2018; Sakar et al., 2019). It contains ten numerical and eight categorical attributes. The numerical attributes record the number of Administrative, Informational and Product-Related pages visited and the time spent on each page

category, together with Bounce Rate, Exit Rate, Page Value and Special Day proximity. Bounce Rate captures the proportion of visitors who leave the site immediately after landing on a page, Exit Rate captures the proportion of page views from which a visitor leaves the site, and Page Value estimates the average monetary value associated with a page relative to completed transactions.

Bounce Rate, Exit Rate and Page Value differ from the other numerical attributes because they are historical page-level averages rather than live measurements of an individual visitor. These values are attached to a session according to the pages that session includes.

Bounce Rate and Exit Rate therefore act as proxies for page “stickiness”. A session containing pages that historically retain visitors tends to differ statistically from one containing pages associated with higher historical abandonment. Page Value goes further by reflecting historical revenue associated with pages and therefore represents commercial value rather than a direct measure of the current visitor's intent. This interpretation is consistent with Satu and Islam (2023).

The categorical attributes record the month of visit, operating system, browser, region, traffic source, visitor type and whether the visit occurred on a weekend, with Revenue (True/False) as the binary class label used for prediction.

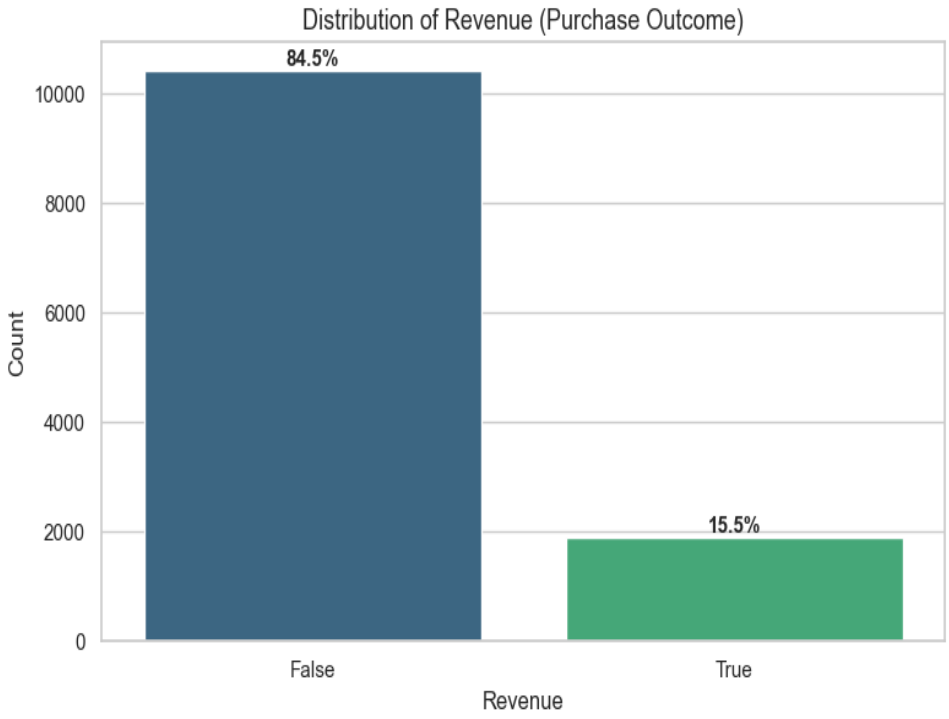


Figure 1. Distribution of the Revenue class label across the 12,330 sessions.

As Figure 1 shows, only 15.5% of sessions resulted in a purchase. This is notably higher than typical industry-wide e-commerce conversion rates, generally reported in the 2–4% range (Speed Commerce, 2026), suggesting that the dataset may exclude very short or low-engagement sessions rather than represent raw, unfiltered site traffic. The imbalance nevertheless has direct implications for model selection and evaluation, discussed in Section 3.

2.3 Predictive Modeling Techniques for Decision-Making

A number of supervised classification algorithms were used to predict a binary outcome, which in this case would be a visitor's purchase intent. Logistic Regression predicts the probability that a session belongs to a given class, based on a weighted combination of the input variables. It remains a popular baseline model because its weights are easy to interpret. A decision-maker can see not just what the model predicts, but which variables are driving that prediction, and in which direction (Hosmer, Lemeshow and Sturdivant, 2013). Decision Trees function differently as they repeatedly split the data into smaller groups, each time choosing the variable that best separates purchasers from non-purchasers. This produces a set of simple, readable rules that one can easily understand but a tree that has grown too deep tends to over-fit, which is when the algorithm memorizes the training data rather than learning general patterns and hence tends to perform poorly with the test data (Breiman, 2001). This weakness is fixed by random forests, which train many decision trees on different random samples of the data and averages their predictions. As more trees are added, the model's error stabilizes, and its overall performance depends both on how accurate each individual tree is and on how different the trees are from one another (Breiman, 2001).

A second group of ensemble methods, known as boosting, takes a different approach. Instead of training many trees independently and averaging them, boosting builds trees one after another, with each new tree specifically trying to fix the mistakes made by the trees before it. AdaBoost, one of the earliest and most influential boosting methods, does this by giving more weight to the examples the model got wrong in the previous round, forcing the next tree to pay closer attention to them (Freund and Schapire, 1997). Gradient Boosting takes this idea further whereby instead of just reweighting mistakes, each new tree is trained to predict the remaining error left by the trees before it (Friedman, 2001).

The next two approaches represent distinct modeling philosophies from those discussed above. Naive Bayes is a probabilistic classifier that applies Bayes' theorem directly by estimating the probability of each class by combining a prior belief with the observed evidence in each feature, under a simplifying assumption that features are conditionally independent given the class (Domingos and Pazzani, 1997). Multilayer Perceptrons (MLPs) take an entirely different approach. They learn through networks of interconnected, weighted nodes that are trained via backpropagation. It adjusts each connection's weight to reduce the difference between the network's output and the true label (Rumelhart, Hinton and Williams, 1986). Unlike the other methods discussed, an MLP can learn complex, non-linear feature interactions automatically, without those interactions needing to be specified in advance.

Choosing an appropriate way to evaluate these models matters just as much as choosing the models themselves. It is clear that purchases are rare compared to non-purchases and judging a model on accuracy alone can be misleading. A model that always guesses "no purchase" would still be right 84.5% of the time despite being completely useless for actually identifying buyers. Using precision, recall, F1-score, and AUC (Area Under the ROC Curve) would give a fuller picture of how well a model actually recognizes the minority group. AUC in particular is useful because it summarizes how well a model separates the two classes across every possible decision threshold, rather than just one fixed cutoff point, making it a fairer way to compare different models against each other (Fawcett, 2006). To reduce the bias toward the majority class caused by the imbalance during training, techniques such as class-weighting or resampling are commonly used. One resampling method that is commonly used is SMOTE, which generates synthetic examples of the minority class to balance the training data (Chawla et al., 2002).

2.4 Mathematical Foundations: Probability, Uncertainty and Utility in Decision-Making

One major thing to note is that the classifiers in Section 2.3 do not predict whether a customer will complete a purchase. They provide a probability, P(Revenue = True | X), conditional on the session's observed features X. Under a threshold of 0.5, a session with probability 0.55 and one with 0.95 are both classified as "likely to purchase," yet they represent very different levels of confidence and should not trigger the same response.

For probabilistic classifiers, the model output can be interpreted as an estimated conditional probability of conversion. Although algorithms derive this estimate differently, Logistic Regression does this through a logistic function (Hosmer, Lemeshow and Sturdivant, 2013), Naive Bayes through Bayes' theorem (Domingos and Pazzani, 1997), random forests through averaged votes (Breiman, 2001), boosted trees through a logistic transformation of their summed predictions (Friedman, 2001), and Multilayer Perceptrons through an output layer (Rumelhart, Hinton and Williams, 1986). The common probabilistic output allows predictions to be compared and linked to a decision threshold.

The dataset's 15.5% conversion rate (Section 2.2) provides the baseline against which individual session probabilities can be interpreted. Naive Bayes applies this logic explicitly, combining a prior with feature likelihoods under a conditional-independence assumption (Domingos and Pazzani, 1997). For the other classifiers, the base rate is contextual as it establishes that a "no purchase" prediction is correct by default 84.5% of the time, which is why accuracy alone is misleading.

A classification threshold is required when converting a probability into a binary action. Typically 0.5 is used but it is not a rule, as it implicitly assumes that false positives and false negatives are equally costly. In e-commerce this is rarely true as a missed buyer represents lost revenue, while an unnecessary intervention represents a wasted discount. Because these costs are asymmetric, the optimal threshold is generally not 0.5.

Added on to this, two sources of uncertainty affect prediction. Aleatoric uncertainty is irreducible randomness in customer behavior, seen when two sessions with identical features differ in outcome due to factors the dataset does not capture. It sets a practical ceiling on achievable accuracy. Epistemic uncertainty, by contrast, arises from the model's limited exposure to the full range of possible behavior and can in principle be reduced with more data, better features, or improved modeling. The gap between achieved and perfect performance reflects both.

Given a predicted probability, expected utility theory (Von Neumann and Morgenstern, 1944) provides a basis for action. If p is the predicted probability of conversion, B the benefit of a successful intervention, and C the cost of an unnecessary one, intervening maximizes expected utility whenever: $p \times B > (1 - p) \times C$, which rearranges to $p^* = C / (B + C)$

This threshold need not equal 0.5. If, for example, B = £40 and C = £4, then $p^* \approx 0.09$ which is far below 0.5, reflecting that intervening is cheap relative to a recovered sale. The framework makes the organization's cost assumptions explicit rather than leaving them hidden in an arbitrary cutoff.

3. Discussion and Analysis

3.1 Data Preparation

The dataset was loaded and checked for completeness. It has 12,330 sessions and 18 attributes, with no missing values. The Boolean fields Weekend and Revenue were cast to binary integers, and the two text-valued categorical attributes, Month and VisitorType, were one-hot encoded using pd.get_dummies, which was preferred to label encoding since neither is ordinal. The coded identifiers (OperatingSystems, Browser, Region and TrafficType) were already stored as integers and were kept as they were. This expanded the feature matrix from 18 original attributes to 28 model inputs.

Numerical attributes were left unscaled for the tree-based and Naive Bayes classifiers, since tree splits are invariant to monotonic transformations. Logistic Regression used standardized features (StandardScaler, fitted on the training set only and then applied unchanged to the test set to avoid leaking test information into training); the MLP was also standardized in the cross-validation described in Section 3.3.

The data was then partitioned into a 75% training set and a 25% held-out test set using stratified sampling (stratify=y), preserving the 84.5% / 15.5% class ratio in both partitions. This matters because the minority class is rare and an unstratified split could by chance produce a materially different class balance in the test set, making model comparisons unreliable.

To address the imbalance, classifiers accepting class weights were trained with class_weight = 'balanced' while AdaBoost, Gradient Boosting and Naive Bayes, which do not expose this parameter, received equivalent weighting via sample_weight (a weight of 5.5 for purchasing sessions). The MLP, which supports neither option, was trained without weighting. Class-weighting was preferred to SMOTE because it adjusts the loss function without altering the training data, avoiding synthetic minority examples whose feature combinations may not correspond to any real customer.

3.2 Exploratory Findings

A correlation analysis of the ten numerical attributes (Figure 2) shows that page counts and their duration counterparts move together, most strongly for Product-Related pages ($r = 0.86$) and more moderately for Administrative ($r = 0.60$) and Informational pages ($r = 0.62$), as would be expected. Bounce Rate and Exit Rate are almost redundant ($r = 0.91$), since both capture aspects of session abandonment. Page Value shows only weak correlation with every other numerical feature ($|r| \leq 0.17$), suggesting that it carries distinct, non-redundant information about purchase likelihood.

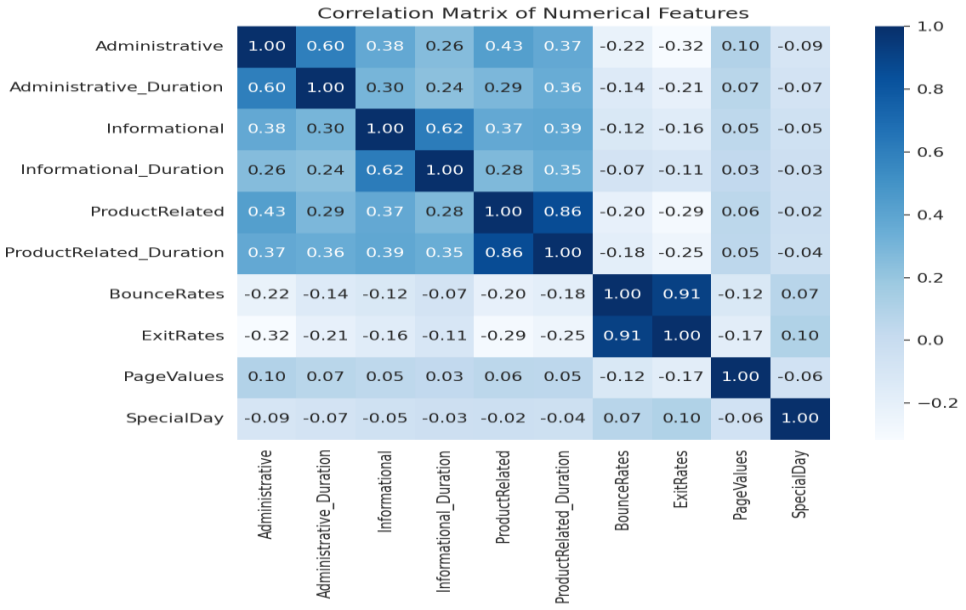


Figure 2. Correlation matrix of the ten numerical attributes.

It is evident that purchasing sessions look clearly different from non-purchasing ones (Figure 3). Most non-purchasing sessions have a Page Value of zero, whereas purchasing sessions have a median of roughly 15–20 and a long upper tail. Bounce Rates move in the opposite direction but support the same conclusion: they sit close to zero for buyers, while non-buyers include many sessions made up of pages with high historical bounce rates. Page Value should be treated with some caution, though. As Section 2.2 explains, it is calculated from the historical revenue of the pages viewed, so it is built from purchase data. If that history includes a session's own purchase, the attribute would give the model part of the answer in advance (a problem known as data leakage), a possibility that cannot be ruled out from the dataset documentation.

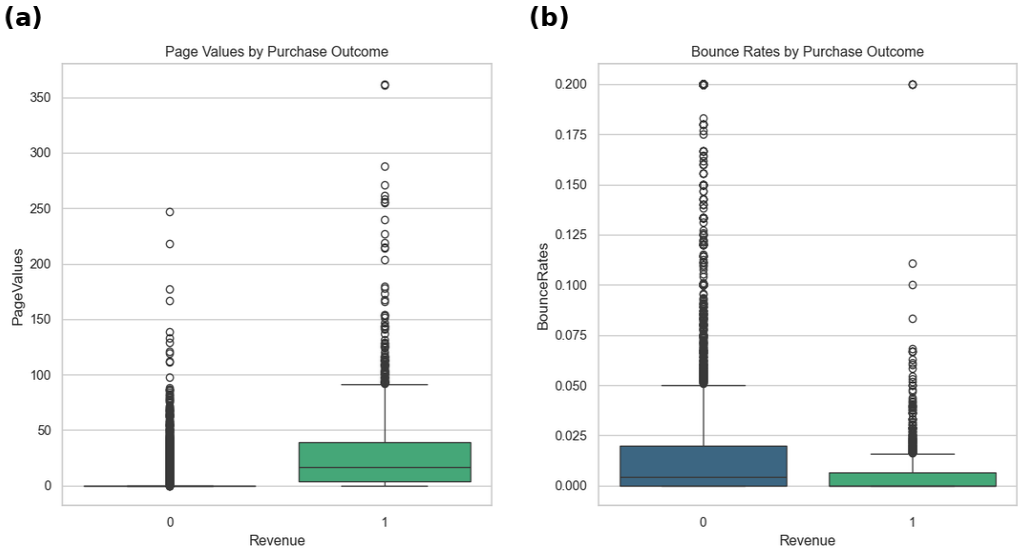


Figure 3. (a) Page Values and (b) Bounce Rates by purchase outcome (Revenue: 0 = no purchase, 1 = purchase).

We can also note that conversion varies sharply with the time of year (Figure 4). It is lowest in February (1.6%), sits at around 10–11% in March, May and June, rises steadily from July (15.3%) to October (20.9%) and peaks in November at 25.4%, plausibly reflecting Black Friday and pre-holiday shopping, before falling to 12.5% in December. The dataset covers only ten months (January and April are missing), so the pattern should not be read as a full annual cycle, and the very low February rate should be treated with caution. It nonetheless suggests that spend and interventions should be timed to the September-November period, when conversion is highest.

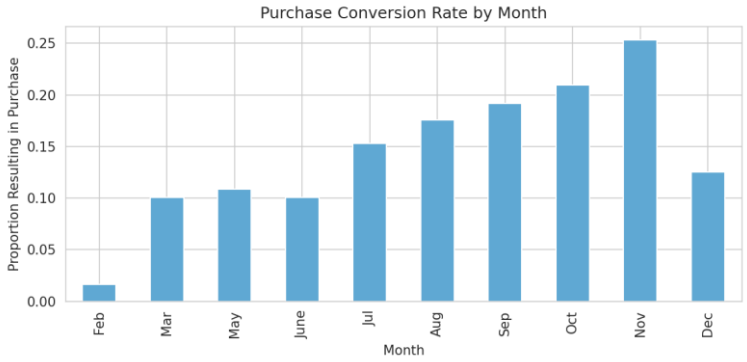


Figure 4. Purchase conversion rate by month of visit.

A less intuitive pattern arises when you consider the visitor types (Figure 5a). Returning visitors make up around 85% of sessions and roughly three-quarters of all purchases, yet they convert at the lowest rate (13.9%), compared with 18.8% for the small “Other” group and 24.9% for new visitors. The bounce rates offer a possible explanation (Figure 5b). Because Bounce Rate is a historical page-level average (Section 2.2), it describes the pages a session included rather than that visitor’s own behavior. New visitors, whether or not they buy, are mostly on pages with very low bounce rates, whereas returning visitors who do not buy often view pages with high bounce rates. This suggests that new visitors tend to arrive on pages built to convert, for example through a campaign, while many returning visitors are browsing or comparing. The data cannot confirm this, but it indicates that visitor type interacts with engagement and should be retained as a predictor.

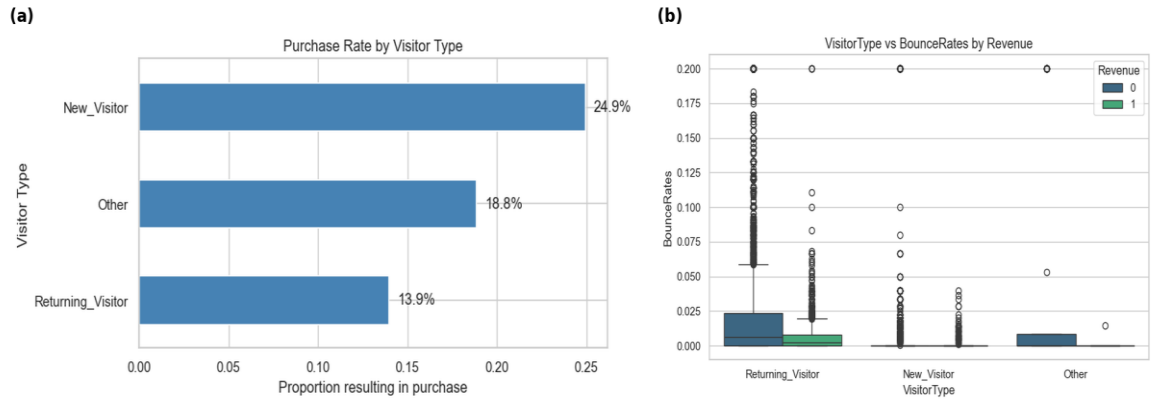


Figure 5. (a) Purchase rate by visitor type and (b) bounce rate by visitor type and purchase outcome.

Taken together, the exploratory analysis suggests that in-session engagement (Page Value and Bounce Rate), time of year and visitor type are all associated with conversion. The high correlation between Bounce Rate and Exit Rate should also be kept in mind when interpreting those results.

3.3 Model Development and Comparison

Seven classification models were trained on 75% of the data (9,247 sessions) and tested on the remaining 3,083 sessions, of which 477 were purchases. The models were Logistic Regression, a Decision Tree, a Random Forest, AdaBoost, Gradient Boosting, Naive Bayes and a Multilayer Perceptron (MLP). Because only 15.5% of sessions end in a purchase, each model was adjusted to give more weight to the purchase class, except the MLP, which does not support this in scikit-learn. Table 1 shows test-set results at the default 0.5 threshold, ordered by AUC. Precision, recall and F1-score refer to purchases.

Model	Accuracy	Precision	Recall	F1-score	AUC
Gradient Boosting	**0.8693**	**0.5499**	**0.8553**	**0.6694**	**0.9345**
Random Forest	0.9046	0.7668	0.5514	0.6415	0.9276
AdaBoost	0.8647	0.5418	0.8155	0.6510	0.9177
Logistic Regression	0.8586	0.5304	0.7505	0.6215	0.9033
MLP	0.8884	0.6572	0.5828	0.6178	0.8951
Decision Tree	0.8440	0.4973	0.7652	0.6028	0.8515
Naive Bayes	0.7139	0.3130	0.7107	0.4346	0.8026

Table 1. Test-set performance of the seven classification models, ordered by AUC.

Gradient Boosting achieved the strongest overall performance, with the highest AUC (0.9345), F1-score (0.6694) and recall (0.8553). The Random Forest had the next-highest AUC (0.9276) and recorded the highest accuracy (0.9046) and precision (0.7668), but its recall was only 0.5514, so it missed almost half of all purchasers.

On the single test split, the difference in AUC between the two is small (0.007), so the comparison was repeated using five-fold stratified cross-validation on the full dataset, with the same settings and class weights (Table 2; the MLP was standardized in this analysis). The ranking by AUC is stable and the gap between Gradient Boosting and the Random Forest narrows further, to 0.9322 against 0.9306 on average. Gradient Boosting did score higher on four of the five folds, but the mean difference (0.0017) is smaller than its variation across folds (standard deviation 0.0022), so the two models should be regarded as equally good at ranking sessions. What separates them, however, is the operating point. At the 0.5 threshold, Gradient Boosting's recall averaged 0.8443 against 0.5419 for the Random Forest, whereas the Random Forest's precision was higher (0.7532 against 0.5388). Gradient Boosting's F1-score was also higher (0.6577 against 0.6298) and more stable across folds (standard deviation 0.0054 against 0.0239).

Model	AUC	F1-score	Recall	Precision
Gradient Boosting	**0.9322 ± 0.0031**	**0.6577 ± 0.0054**	**0.8443 ± 0.0148**	**0.5388 ± 0.0081**
Random Forest	0.9306 ± 0.0038	0.6298 ± 0.0239	0.5419 ± 0.0317	0.7532 ± 0.0274
AdaBoost	0.9185 ± 0.0030	0.6509 ± 0.0164	0.8145 ± 0.0083	0.5423 ± 0.0220
Logistic Regression	0.9043 ± 0.0049	0.6148 ± 0.0041	0.7641 ± 0.0077	0.5143 ± 0.0044
MLP	0.8825 ± 0.0110	0.5763 ± 0.0192	0.5723 ± 0.0312	0.5825 ± 0.0368
Decision Tree	0.8678 ± 0.0171	0.6177 ± 0.0177	0.7914 ± 0.0289	0.5068 ± 0.0170
Naive Bayes	0.8174 ± 0.0067	0.4583 ± 0.0050	0.7537 ± 0.0169	0.3294 ± 0.0050

Table 2. Five-fold stratified cross-validation results (mean ± standard deviation across folds), ordered by AUC.

In such a dataset, accuracy can be misleading. A model that predicted “no purchase” for every session would already be 84.5% accurate, so the Random Forest’s 90.5% represents a gain of only about six percentage points. Among the simpler models, Logistic Regression (AUC 0.9033) outperformed both the single Decision Tree (0.8515) and the MLP (0.8951), suggesting that much of the signal can be captured by a near-linear decision boundary and that the ensemble methods add a modest improvement (Figure 6). In contrast to Sakar et al. (2019), the MLP did not outperform the ensemble methods here (AUC 0.8951 against 0.9345 for Gradient Boosting), although it was trained without class weighting or tuning.

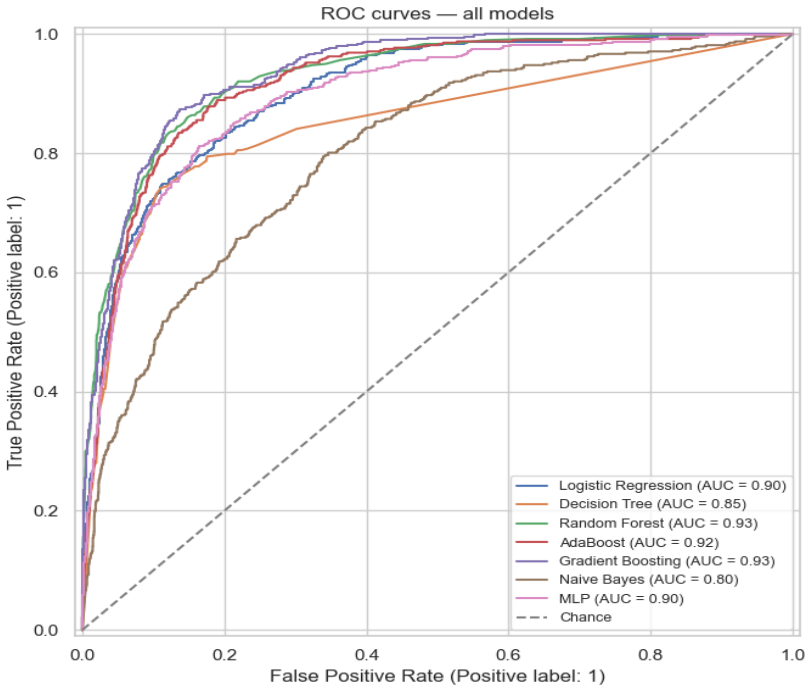


Figure 6. ROC curves for the seven classifiers on the test set.

The results separate into two profiles, which guide the choice of model.

1. Low-cost approach: Gradient Boosting (recall 0.8553) and AdaBoost (0.8155) favor recall as they catch most purchasers but flag many non-purchasers. If the priority is to avoid missing sales opportunities, for example by offering a low-cost incentive to any visitor showing signs of purchase intent, these models are preferable, since the cost of an unnecessary offer is typically lower than the cost of a missed sale.
2. High-cost approach: Random Forest (precision 0.7668) and the MLP (0.6572) favor precision and suit scarce or expensive interventions, such as live customer-service agents, where acting on false positives is costly.

Gradient Boosting is therefore adopted as the primary model, given its highest F1-score and recall and an AUC essentially equal to the Random Forest's, with the Random Forest as the alternative where precision matters more.

3.4 Error Analysis and Feature Importance

Figure 7 shows the confusion matrix for the Gradient Boosting model on the test set. Of the 477 sessions that ended in a purchase, 408 were correctly identified and 69 were missed. Of the 2,606 non-purchasing sessions, 2,272 were correctly classified and 334 were incorrectly flagged as likely purchases. A flagged session is therefore a genuine purchaser only about 55% of the time (precision 0.5499), so almost half of interventions would be directed at someone who would not have bought.

By comparison, the Random Forest found 263 purchasers and raised 80 false alarms, whereas Gradient Boosting found 145 more purchasers (408) at the cost of 254 more false alarms (334), or about 1.75 additional false alarms for every extra purchaser identified. Across the cross-validation folds (Table 2), Gradient Boosting's recall averaged 0.8443 and its precision 0.5388, compared with 0.5419 and 0.7532 for the Random Forest, so the pattern is not specific to one split. That trade is favorable when interventions are cheap and less so when each one is costly.

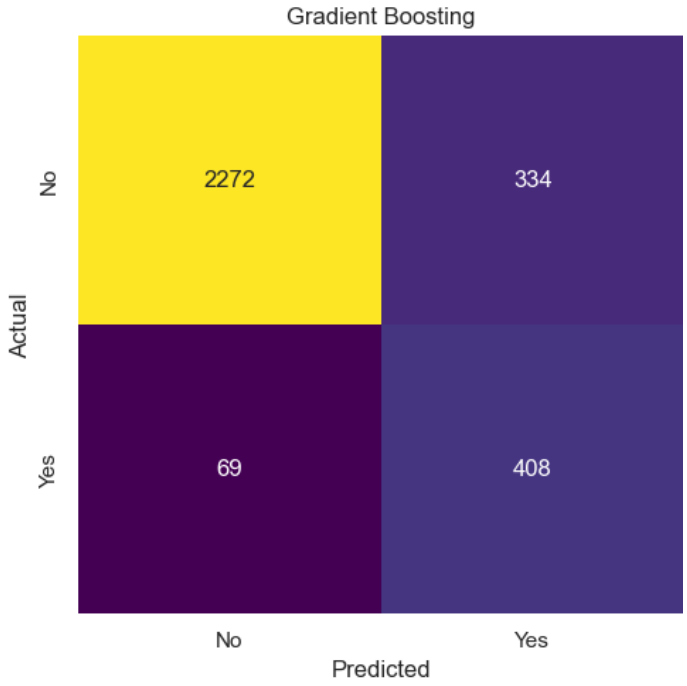


Figure 7. Confusion matrix for the Gradient Boosting model on the test set.

Figure 8 shows which variables the Gradient Boosting model uses most when predicting purchases. Both measures show that Page Value is by far the most important variable. In the impurity-based measure, it accounts for about 77% of the importance. The permutation measure gives a similar result: shuffling the values of Page Value reduces test-set accuracy by about 17 percentage points. Other variables, including Product-Related Duration, Administrative page count, Bounce Rate and new-visitor status, have much smaller effects. Browser, Region and Operating System have very little influence on the predictions.

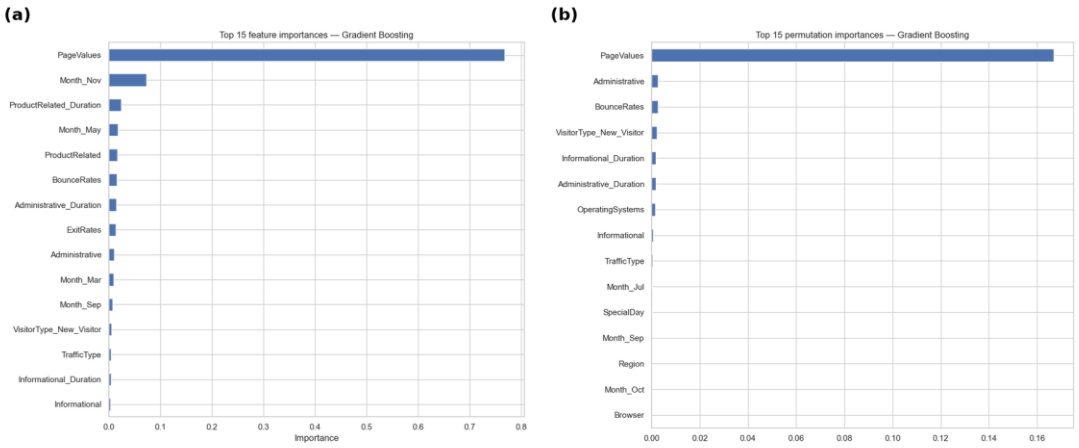


Figure 8. Gradient Boosting feature importance: (a) impurity-based importance and (b) permutation importance on the test set (top 15 attributes).

The two measures therefore agree that Page Value is the main source of predictive information but since Page Value is related to the value of pages viewed during a session, it may contain information closely linked to the purchase outcome. The model should therefore be tested again without Page Value, or using only information available earlier in the session, to check whether its performance remains strong.

If Page Value can be calculated while a visitor is still browsing, the model could also be used in real time. It could identify visitors who appear likely to purchase and allow the organization to respond before the session ends, for example by showing a targeted offer or providing live-chat support.

3.5 Uncertainty, Probability Calibration and Expected Utility

The conventional 0.5 classification threshold is an arbitrary default rather than a business-optimal choice, and where to draw the line between “intervene” and “do not intervene” should reflect the relative costs and benefits at stake rather than a statistical convention. Figure 9 shows how Gradient Boosting's precision, recall and F1-score change with the threshold.

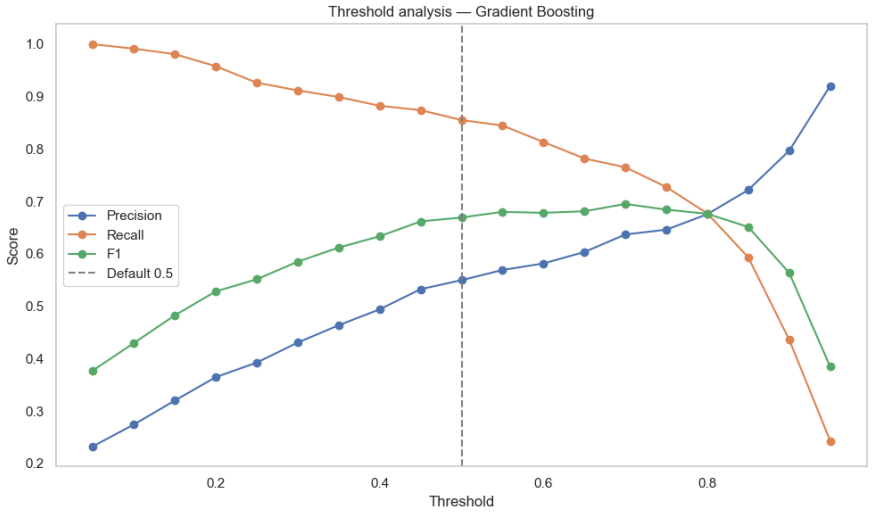


Figure 9. Precision, recall and F1-score of the Gradient Boosting model at different classification thresholds.

Applying the expected-utility rule from Section 2.4 requires model scores that behave like probabilities, and Gradient Boosting's scores do not. Because purchasing sessions were weighted 5.5 times more heavily during training, the model's scores run higher than true probabilities, so a score of 0.5 does not mean a 50% chance of purchase.

Consider an illustrative case in which a successful intervention, such as a personalized discount, recovers a margin of B = £40, while an unnecessary one costs C = £4. The utility-maximizing probability threshold is $p^* = C / (B + C) \approx 0.09$. Allowing for the extra weight given to purchasers in training, a session with a true 9% chance of purchase receives a score of about 0.35, so 0.35 is the cut-off to apply to the model's scores. At 0.35, Figure 9 indicates recall of roughly 0.90 and precision of roughly 0.46, against 0.86 and 0.55 at the default 0.5. If the intervention were costly, such as staff-arranged free shipping at C = £20, the score cut-off would rise to about 0.73, where Figure 9 shows precision of roughly 0.65 and recall of roughly 0.75.

Cheap interventions therefore justify a lower cut-off and higher recall, and costly ones a higher cut-off and higher precision. Because the scores are not true probabilities, any chosen cut-off should be piloted before wider rollout.

4. Conclusion and Recommendations

This report set out to build a predictive model of purchasing intent from session-level browsing data and to connect its outputs to business decisions. The exploratory data analysis showed that the data is heavily imbalanced (84.5% non-purchasing sessions), that purchasers have higher Page Values and lower bounce rates, and that conversion peaks in November (25.4%) and is higher for new visitors (24.9%) than for returning ones (13.9%). Of the seven models compared, Gradient Boosting performed best overall (AUC 0.9345, recall 0.8553), but cross-validation showed that it ranks sessions no better than the Random Forest. The two differ mainly in how they balance recall against precision. Page Value dominated the feature importance rankings, so the model depends heavily on a single attribute. Because purchasers were weighted more heavily in training, the model's scores are not true probabilities. Once this is corrected for, the appropriate decision threshold depends on the cost of the intervention (about 0.35 for a cheap discount and 0.73 for costly shipping) rather than defaulting to 0.5.

Five recommendations follow:

- Deploy a scoring layer based on Gradient Boosting, using the Random Forest instead where precision matters more, after re-testing without Page Value to confirm that performance does not rely on information unavailable at the time of the decision.
- Set thresholds according to the cost of each intervention (around 0.35 for low-cost actions and 0.7 or higher for costly ones), and pilot them before wider rollout.
- Audit page-level content on high-traffic product pages, since Page Value was the most influential attribute.
- Concentrate acquisition spend ahead of the November peak, and use low-conversion months such as February and March for retention campaigns aimed at returning visitors, who convert less often than new visitors (Figures 4 and 5).
- Monitor and periodically retrain the model, checking that its scores remain well calibrated (Chapman et al., 2000).

The study has some limitations. It uses a single one-year dataset (covering ten months) from one organization, and the models were not tuned. The model's scores also run higher than true probabilities, so the threshold values are only approximate and should be checked before deployment. In addition, Page Value may partly reflect the purchase outcome, so the results may be somewhat optimistic. Future work could use sequence models, such as the LSTM networks evaluated by Sakar et al. (2019), to learn from the order in which pages are visited. Overall, a small set of behavioral signals, linked to the cost of each action, can support a useful decision tool, once the reliance on Page Value and the accuracy of the scores have been checked.

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